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Radiation imaging apparatus, radiation detector, method and apparatus for controlling radiation imaging apparatus, and program for the same

Abstract

A setting unit configured to set a reference orientation for calculating the orientation of a radiation detector and a reference orientation for calculating the orientation of the radiation generator sets the reference orientations of the radiation detector and the radiation generator in response to determining that a mobile radiography unit equipped with the radiation generator stands still, with the radiation detector housed, to determine whether the radiation generator and the radiation detector are arranged according to the imaging procedure.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a Continuation of International Patent Application No. PCT/JP2022/015490, filed Mar. 29, 2022, which claims the benefit of Japanese Patent Application No. 2021-058489, filed Mar. 30, 2021, and No. 2022-040251, filed Mar. 15, 2022, all of which are hereby incorporated by reference herein in their entirety.

TECHNICAL FIELD
(1) The present invention relates to a radiation imaging apparatus, a radiation detector, a method and an apparatus for controlling the radiation imaging apparatus, and a program for the same.
BACKGROUND ART
(2) At present, flat panel detectors (FPDs) made of a semiconductor material are in widespread use as radiation detectors for use in medical diagnostic imaging and nondestructive examinations using radiation such as X rays. Radiation imaging apparatuses are in use in which such a radiation detector and a radiation generator or the like are combined.
(3) As a function of such a radiation imaging apparatus, a function for supporting positioning of the

radiation field surface of radiation emitted from a radiation generator and the incident surface of the radiation detector by calculating the orientations of the radiation generator and the radiation detector and displaying the orientations is in practical use.
(4) An example of a method for calculating the orientations of the radiation generator and the radiation detector is calculating the orientations, with an acceleration sensor or a gyroscope provided at each of the radiation generator and the radiation detector, from accelerations, which are output values from the acceleration sensors, or angular velocities, which are output values from the gyroscopes.
(5) For example, the calculation of the orientation using the gyroscope is made by cumulating (integrating) the angular velocities in a minute time obtained by the gyroscope. The calculation of the orientation using the acceleration sensor is made by cumulating accelerations obtained by the acceleration sensor to calculate a speed at a certain time and further cumulating the speeds again to convert it to a displacement (position).
(6) However, what can be calculated by the above method is the sum of the orientation changes, and the current accurate orientation cannot be calculated unless a reference orientation at a certain time (hereinafter, referred to as reference orientation) is known.
(7) To avoid such a problem, PTL 1 discloses a radiation imaging apparatus including an input unit for setting a reference orientation and a positioning ruler, which is a structure for positioning. The user of the radiation imaging apparatus can set the reference orientation by providing an instruction for setting the reference orientation (calibration processing) via the input unit when the radiation detector abuts on the positioning ruler.
(8) For example, for supporting the positioning of a radiation generator and a radiation detector, PTL 2 discloses a technique for setting vectors based on the orientation information on a radiation generator and radiation detectors in an environment in which a plurality of radiation detectors is used. This allows selection of a radiation detector whose vector forming with the vector of the radiation generator is less than a predetermined value.
CITATION LIST
Patent Literature
(9) PTL 1 Japanese Patent Laid-Open No. 2018-007923
(10) PTL 2 US Patent Application, Publication No. 2015/0098551
(11) However, the radiation imaging apparatus disclosed in PTL 1 requires the user to provide an instruction via the input unit when setting the reference orientation, complicating the procedure for use.
(12) In some medical front, the radiation detector is used in a bag from the aspect of good hygiene. This makes it difficult for the user to visually determine the front and the back of the radiation detector. In applying the technique of PTL 2 in such a situation, when a notification that the angle of the radiation generator and the angle of the radiation detector are mismatched is given, it cannot be determined whether the angles are mismatched or the angles are matched but the back of the radiation detector faces. If the user visually determines that the orientations are coincide and applies radiation, with the back of the radiation detector facing, an image not suitable for diagnosis is formed.
SUMMARY OF INVENTION
(13) A first aspect of the present invention provides a radiation imaging apparatus configured to set reference orientations for calculating the orientations of the radiation detector and the radiation generator without the user, such as an operator, performing complicated operations.
(14) A second aspect of the present invention provides a radiation imaging apparatus configured to notify the user whether the radiation detector is positioned relative to the radiation generator according to the imaging procedure in matching the angles using the orientation information on the radiation generator and the radiation detector.
(15) A radiation imaging apparatus according to the first aspect includes a radiation detector configured to detect radiation, a mobile radiography unit configured to house the radiation detector and including a radiation generator, and a setting unit configured to set at least one of a reference orientation of the radiation detector for calculating an orientation of the radiation detector and a reference orientation of the radiation generator for calculating an orientation of the radiation generator, wherein the radiation imaging apparatus is configured to perform radiation imaging to generate an image based on the radiation, and wherein the setting unit performs the setting in response to determining that the mobile radiography unit

stands still, with the radiation detector housed in the mobile radiography unit.
(16) A radiation imaging apparatus according to the second aspect includes a radiation detector configured to detect radiation, a mobile radiography unit configured to house the radiation detector and including a radiation generator, an orientation acquisition unit configured to acquire relative relationship between an orientation of the radiation generator and an orientation of the radiation detector, a first orientation measuring unit configured to measure information on the orientation of the radiation generator, and a second orientation measuring unit configured to measure information on the orientation of the radiation detector, wherein the orientation acquisition unit acquires the relative relationship from an angle formed between a first normal vector and a second normal vector, the first normal vector being aligned with a direction of the radiation from the radiation generator and being based on a value measured by the first orientation measuring unit, the second normal vector being perpendicular to a detector plane of the radiation detector and being based on a value measured by the second orientation measuring unit.
(17) Using a radiation imaging apparatus according to an embodiment of the present invention allows setting reference orientations for calculating the orientations of the radiation detector and the radiation generator without the need for a user, such as an operator, to perform complicated operations. This also allows notifying the user of the result of determination of the arrangement of the radiation generator and the radiation detector in positioning the radiation detector and the radiation generator.
(18) Further features of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) FIG. **1**A is a diagram illustrating a configuration example of a radiation imaging apparatus and a radiation generator according to a first embodiment of the present invention.
(2) FIG. **1**B is a diagram illustrating a configuration example of the radiation imaging apparatus and the radiation generator according to the first embodiment.
(3) FIG. **2** is a diagram illustrating a mobile radiography unit and a flat panel detector (FPD) according the first embodiment.
(4) FIG. **3**A is a diagram illustrating the relationship among the coordinates of the FPD and the coordinates of the radiation generator according to the first embodiment.
(5) FIG. **3**B is a diagram illustrating the relationship among the coordinates of the FPD and the coordinates of the radiation generator according to the first embodiment.
(6) FIG. **3**C is a diagram illustrating the relationship among the coordinates of the FPD and the coordinates of the radiation generator according to the first embodiment.
(7) FIG. **4** is a flowchart for the processing of the radiation imaging apparatus according to the first embodiment.
(8) FIG. **5**A is a diagram illustrating a method for initial setting according to the first embodiment.
(9) FIG. **5**B is a diagram illustrating a method for initial setting according to the first embodiment.
(10) FIG. **5**C is a diagram illustrating a method for initial setting according to the first embodiment.
(11) FIG. **5**D is a diagram illustrating a method for initial setting according to the first embodiment.
(12) FIG. **6**A is a diagram illustrating an example of a method for examining the arrangement of the radiation generator and the FPD according to the first embodiment.
(13) FIG. **6**B is a diagram illustrating an example of a method for examining the arrangement of the radiation generator and the FPD according to the first embodiment.
(14) FIG. **6**C is a diagram illustrating an example of a method for examining the arrangement of the radiation generator and the FPD according to the first embodiment.
(15) FIG. **7** is a diagram illustrating a mobile radiography unit and an FPD according a modification.
(16) FIG. **8** is a diagram illustrating a configuration example of a radiation imaging apparatus according to a second embodiment.
(17) FIG. **9** is a flowchart for the processing of the radiation imaging apparatus according to the second embodiment.
(18) FIG. **10** is a flowchart for the processing of a radiation imaging apparatus according to a third

embodiment.
(19) FIG. **11**A is a diagram illustrating the details of a method for moving a radiation generator according to the third embodiment.
(20) FIG. **11**B is a diagram illustrating the details of a method for moving the radiation generator according to the third embodiment.
(21) FIG. **11**C is a diagram illustrating the details of a method for moving the radiation generator according to the third embodiment.
(22) FIG. **12** is a diagram illustrating a configuration example of a radiation imaging apparatus according to a fourth embodiment.
DESCRIPTION OF EMBODIMENTS
(23) The present invention will be described hereinbelow using exemplary embodiments with reference to the accompanying drawings. It is to be understood that the following embodiments do not limit the invention according to the claims. Although the embodiments describe a plurality of characteristics, not all of the plurality of characteristics is absolutely necessary for the invention and may be freely combined. In the accompanying drawings, like or similar components are given like reference signs, and redundant descriptions will be omitted. The term “radiation” is typically X-rays, but may include α rays, β rays, γ rays, particle rays, and cosmic rays.
First Embodiment
(24) FIG. **1**A is a conceptual diagram illustrating a configuration example of a radiation imaging apparatus **100** according to a first embodiment of the present invention.
(25) A mobile radiography unit **101** is a movable truck with wheels or the like, on which a radiation generator **102** is mounted. The mobile radiography unit **101** is used for radiation imaging together with a flat panel detector (FPD)**103** (an example of a radiation detector). The FPD **103** is disposed on the back of a subject **105** on a bed **104** for radiation imaging. The FPD **103** generates a radiological image of the subject **105** based on the radiation radiated from the radiation generator **102** and applied to the radiation detecting unit **200** through the subject **105**.
(26) The radiation detecting unit **200** includes a scintillator that converts the radiated radiation to light and a pixel array in which photoelectric conversion elements that convert light from the scintillator to electrical charge are disposed in a two-dimensional array. The electrical charge based on the radiation, generated by the radiation detecting unit **200**, flows to a read circuit (not shown) in sequence by scanning the pixel array with a drive circuit (not shown) to generate an image based on the radiation. The image generated by the radiation detecting unit **200** is sent to a second control unit **203** and is processed as necessary by the second control unit **203**.
(27) The mobile radiography unit **101** further includes, in addition to the radiation generator **102**, a first arm **106** and a second arm **107** that support the radiation generator **102**. The mobile radiography unit **101** further includes a casing **108**, a base **109**, and an FPD housing unit **110**.
(28) In this embodiment, the first arm **106** connects to the radiation generator **102** and the second arm **107**, and the second arm **107** connects to the first arm **106** and the base **109**. The casing **108** has a first display **111** fitted on the outside and contains a battery (not shown), a high-voltage generator (not shown), a first control unit **201** shown in FIG. **2** (not shown in FIGS. **1**A and **1**B), and a first information transmitter and receiver **202** (not shown in FIGS. **1**A and **1**B).
(29) The FPD housing unit **110** is a pocket provided at the mobile radiography unit **101** to mount the FPD **103** on the mobile radiography unit **101**. The FPD housing unit **110** includes a first housing determination unit **112** for determining whether the FPD **103** is housed. Connecting the first housing determination unit **112** and a second housing determination unit **114**, described later, allows a determination whether the FPD **103** is housed in the FPD housing unit **110**.
(30) The FPD housing unit **110** serves as a reference for calculating the position of the FPD **103**, described later, so that the FPD housing unit **110** has a positioning mechanism for positioning the FPD **103**. If the first housing determination unit **112** and the second housing determination unit **114**, described later, are connected by an electrical connector, the electrical connector may also serve as a positioning mechanism.
(31) The first display **111** is formed of a touch panel so as to accept user's input. Other examples of the input device that accepts user's input include a keyboard, a mouse, a speech recognition device, and a

user's posture recognition device using a distance sensor. The FPD **103** has a built-in battery (not shown) so as to be operated using the electric power of the battery.

(32) The FPD **103** of this embodiment includes a second orientation measuring unit **118**, the second housing determination unit **114** provided on the FPD **103** side, a second control unit **203** (not shown in FIGS. **1**A and **1**B) and a second information transmitter and receiver **204** (not shown in FIGS. **1**A and **1**B). The second orientation measuring unit **118** in this embodiment is an acceleration sensor. The orientation measuring unit **118** may be an angular velocity sensor or a geomagnetic sensor.

(33) In FIG. **1**A, the FPD **103** is disposed on the back of the subject **105** but. When the mobile radiography unit **101** is conveyed together with the FPD **103**, the FPD **103** is housed in the FPD housing unit **110** and conveyed. When the user performs radiation imaging, the FPD **103** held in the FPD housing unit **110** is taken out to the back of the subject **105**.

(34) The details of the radiation generator **102** are shown in FIG. **1**B. The radiation generator **102** includes a tube **115** and an aperture **116**. In this embodiment, the second display **117** and the first orientation measuring unit **113** are mounted on the aperture **116**. The first orientation measuring unit **113** of this embodiment is an acceleration sensor. The orientation measuring unit **118** may be an angular velocity sensor or a geomagnetic sensor.

(35) The aperture **116** includes a mechanism (including a stepping motor) that operates in accordance with a command of a computer mounted in the casing **108** to change the size of the radiation field and the rotation angle of the FPD **103**.

(36) Examples of the first housing determination unit **112** and the second housing determination unit **114** are electrical connectors for the mobile radiography unit **101** and the FPD **103**. These electrical connectors are disposed at the mobile radiography unit **101** to charge the FPD **103**. In this embodiment, the electrical connectors are also used to determine that the FPD **103** is housed in the FPD housing unit **110** of the mobile radiography unit **101** when the connectors of the mobile radiography unit **101** and the FPD **103** are connected so that the FPD **103** is being charged.

(37) In this embodiment, when the FPD **103** is housed, the mobile radiography unit **101** and the FPD **103** are electrically connected with the first housing determination unit **112** and the second housing determination unit **114**. However, the first housing determination unit **112** and the second housing determination unit **114** of this embodiment are provided to indicate that the FPD **103** is at a specific position, and therefore do not necessarily have to be electrically connected.

(38) For example, the functions of the first housing determination unit **112** and the second housing determination unit **114** may be achieved by a proximity determination unit, such as Bluetooth® or near-field communication (NFC).

(39) Alternatively, a weight scale for measuring the weight of the FPD **103** may be mounted in the FPD housing unit **110** of the mobile radiography unit **101**. This allows determination that the FPD **103** is housed when the value of the weight scale is approximately the same as the weight of the FPD **103**.

(40) In this case, if the imaging plane of the FPD **103** is rectangular, so that the orientation has to be determined, the orientation is determined using, for example, a gravity sensor mounted on the FPD **103**. As another alternative, a camera is mounted at the mobile radiography unit **101**, and when the camera recognizes the FPD **103**, it is determined that the FPD **103** is housed.

(41) The FPD **103** may be provided with a sensor for determining whether the FPD **103** is housed. For example, sensors that distinguish between light and dark may be provided at the four corners of the FPD **103**. When specific two sensors detect light and the other two sensors detect dark, it can be determined that the FPD **103** is housed in the FPD housing unit **110**. This also allows determination of the orientation of the housing. Furthermore, in the case of a configuration in which the second control unit **203** of the FPD **103** has the function of a setting unit **205**, described later, the determination of mounting can be made only at the FPD **103**, thereby facilitating implementation of the mobile radiography unit **101**.

(42) Referring next to FIG. **2**, a configuration example of the mobile radiography unit **101** and the FPD **103** in this embodiment will be described.

(43) First, the components of the mobile radiography unit **101** will be described. The mobile radiography unit **101** houses a first control unit **201**. The first control unit **201** may be a general-purpose computer constituted by hardware including a central processing unit (CPU), a main storage, such as a dynamic random-access memory (DRAM), and an auxiliary storage, such as a solid state drive (SSD) or a hard

disk drive (HDD). The first control unit **201** has the functions of the setting unit **205** and a reference calculation unit **208**, described later.
(44) The first control unit **201** connects to the radiation generator **102** to control radiation, such as whether to emit radiation. The first control unit **201** connects to the first orientation measuring unit **113** to calculate the reference orientation of the radiation generator **102**. The reference orientation of the radiation generator **102** can be calculated by the reference calculation unit **208** of the first control unit **201** from a value obtained by the first orientation measuring unit **113**.
(45) The first control unit **201** connects to the second orientation measuring unit **118** via the first information transmitter and receiver **202** and the second information transmitter and receiver **204** to calculate the reference orientation of the FPD **103**. The reference orientation of the FPD **103** can be calculated by the reference calculation unit **208** of the first control unit **201** from the value of the second orientation measuring unit **118**. The first control unit **201** determined the timing of setting the reference orientations of the radiation generator **102** and the FPD **103**. This determination is made with the function of the setting unit **205** of the first control unit **201**.
(46) The reference orientation is a reference orientation at a certain time for calculating the orientation of the apparatus. In this embodiment, the radiation generator **102** and the FPD **103** each include an acceleration sensor for calculating the respective orientations.
(47) To calculate the orientation using the acceleration sensor, accelerations obtained by the acceleration sensor are cumulated to calculate a velocity at a certain time and the velocities are cumulated again to be converted to displacement (position). However, what can be calculated with the above method is the sum of the orientation changes and therefore cannot be calculated accurately unless the reference orientation is known. The setting unit **205** determines appropriate timing to set the reference orientations of the radiation generator **102** and the FPD **103**. This timing will be described later.
(48) The first control unit **201** connects to the first information transmitter and receiver **202**. Examples of the first information transmitter and receiver **202** include a wireless local area network (LAN) device, a Bluetooth device, and an ultra wide band (UWB) device. Transmission and reception of information to/from the FPD **103** can be performed by connecting to the second information transmitter and receiver **204** of the FPD **103** using the first information transmitter and receiver **202**.
(49) For example, in this embodiment, the setting unit **205** is mounted in the mobile radiography unit **101**, so that information on the setting of the reference orientation is transmitted to the FPD **103** using the first information transmitter and receiver **202**. The first information transmitter and receiver **202** also has the function of receiving orientation information obtained by the orientation acquisition unit **209** of the FPD **103**. The second information transmitter and receiver **204** connects to the first information transmitter and receiver **202** by radio or the like to be used also for communicating with the mobile radiography unit **101** to change information on radiation control, such as whether to emit radiation, and transmitting images acquired by the FPD **103**.
(50) The first control unit **201** connects also to the first display **111** also serving as an input device, so that the user can input protocol information on radiation imaging via the first display **111**. The first control unit **201** detects input of the protocol information.
(51) In this embodiment, the FPD **103** includes the second control unit **203**, the second information transmitter and receiver **204**, the second orientation measuring unit **118**, and the second housing determination unit **114**. As shown in FIG. **2**, these units are electrically connected to one another. The second control unit **203** can be constituted by a CPU, a main memory, and an auxiliary storage as is the first control unit **201**.
(52) The second control unit **203** may be simpler than the first control unit **201** of the mobile radiography unit **101** because the FPD **103** needs to be compact, lightweight, and power saving. To meet such requirement, the second control unit **203** of the FPD **103** may be a field programmable gate array (FPGA) or a dedicated integrated circuit (IC).
(53) In this embodiment, the function of the orientation acquisition unit **209** is achieved by the second control unit **203**. The orientation acquisition unit **209** calculates the orientation of the FPD **103** from the reference orientation calculated by the reference calculation unit **208** and set by the setting unit **205** and a value calculated by the second orientation measuring unit **118**.
(54) The relationship between an angle θ(t) and an angular velocity ω.sub.θ(t) about X-axis at time t, the

relationship between a rotation angle ω.sub.ϕ(t) and an angular velocity ω.sub.η(t) about Y-axis at time t, and the relationship between a rotation angle η(t) and an angular velocity ϕ(t) about Z-axis at time t is expressed as:

(55) $(t) = \quad (0) + .\mathrm{Math}._{k=1}^{n} \quad (k \quad t) \quad t$ Exp. 1 $\quad (t) = \quad (0) + .\mathrm{Math}._{k=1}^{n} \quad (k \quad t) \quad t$ Exp. 2

$(t) = \quad (0) + .\mathrm{Math}._{k=1}^{n} \quad (k \quad t) \quad t$ Exp. 3

where Δt, is the angular velocity measurement time interval, and n is the number of measurements (t=nΔt).

(56) Using a gyroscope as the second orientation measuring unit **118** allows obtaining the value of an angular velocity in the coordinate system of the gyroscope. In calculating an angle from the angular velocity obtained by the gyroscope, the angular velocity in the coordinate system of the gyroscope is converted to an angle in a desired coordinate system. Known method may be used for this conversion.

(57) The position is derived by calculating the sum of the accelerations to derive the velocity and then summing the velocities. Let x(t), v.sub.x(t), and a.sub.x(t) be X-axis components of position, velocity, an acceleration at time t, respectively, y(t), v.sub.y(t), and a.sub.y(t) be Y-axis components, and z(t), v.sub.z(t), and a.sub.z(t) be Z-axis components, respectively. The velocities v.sub.x(t), v.sub.y(t), and v.sub.z(t) at time t are expressed as:

(58) $v_x(t) = v_x(0) + .\mathrm{Math}._{k=1}^{n} a_x(k \quad t) \quad t$ Exp. 4 $\quad v_y(t) = v_y(0) + .\mathrm{Math}._{k=1}^{n} a_y(k \quad t) \quad t$ Exp. 5

where Δt is the measurement time interval, and n is the number of measurements (t=nΔt) as is the angular velocities (t=nΔt).

(59) Therefore, the positions x(t), y(t), and z(t) are expressed as:

(60) $x(t) = x(0) + .\mathrm{Math}._{k=1}^{n} v_x(k \quad t) \quad t$ Exp. 7 $\quad y(t) = y(0) + .\mathrm{Math}._{k=1}^{n} v_y(k \quad t) \quad t$ Exp. 8

$z(t) = z(0) + .\mathrm{Math}._{k=1}^{n} v_z(k \quad t) \quad t$ Exp. 9

(61) Exps. 1 to 9 may adopt a known numerical integration scheme, such as a trapezoid formula or Simpson's formula, to increase the accuracy of summing. In this case, the accelerations a.sub.x(t), a.sub.y(t), and a.sub.z(t) detected by the acceleration sensor contain a gravity component. For this reason, the direction of gravity exerted on the FPD **103** may be calculated using the angles θ(t), ϕ(t), and η(t) obtained with the gyroscope, and the x, y, and z components of the gravitational acceleration may be subtracted from the accelerations.

(62) FIGS. **3**A to **3**C illustrate the relationship among the coordinates X.sub.D, Y.sub.D, and Z.sub.D of the FPD **103** and the coordinates X.sub.S, Y.sub.S, and Z.sub.S of the radiation generator **102** in this embodiment.

(63) The coordinates of the FPD **103** and the coordinates of the radiation generator **102** are each defined as a coordinate relative to a reference point O. In this embodiment, the reference point O is defined as one point of a portion where the first housing determination unit **112** and the second housing determination unit **114** are connected. For example, one point of a portion where the first housing determination unit **112** and the second housing determination unit **114**, for example, a corner or the center of gravity, is defined as the reference point O.

(64) For the coordinates of the FPD **103**, the axis parallel to and opposite to the gravity is set as the Z-axis, and an X-Y plane is set on the horizontal plane (a plane perpendicular to the direction in which gravity acts) with reference to the center P.sub.D of the FPD **103** (for example, the center of gravity). The y-direction is defined as being parallel to the base **109**, and the x-direction is defined as be perpendicular to the y-direction (and in an X-Y plane).

(65) For the coordinates of the radiation generator **102** as well, the axis parallel to and opposite to the gravity is set as the Z-axis, and an X-Y plane is set on the horizontal plane (a plane perpendicular to the direction in which gravity acts) with reference to the center P.sub.4 of an emission surface. The y-direction is defined as being parallel to the base **109**, and the x-direction is defined as being perpendicular to the y-direction. Calculation of the source image receptor distance (SID), which here refers to the distance to the incident surface of the FPD, uses the position P.sub.3 of the focal point of the tube **115**,

instead of P.sub.4.
(66) The coordinates of the FPD **103** are expressed as vector OP.sub.D, and the orientation of the FPD **103** is defined as rotation about the respective axes X.sub.D, Y.sub.D, and Z.sub.D. If the FPD **103** is determined to be housed in the first housing determination unit **112** of the mobile radiography unit **101**, the orientation of the FPD **103** is determined only from the geometric arrangement of the housed FPD **103** and the FPD housing unit **110**.
(67) In other word, when the FPD **103** is housed in the FPD housing unit **110** and is connected by the first housing determination unit **112** and the second housing determination unit **114**, the vector OP.sub.D representing the orientation and the rotation angles about X.sub.D, Y.sub.D, and Z.sub.D take values that depend only on their respective outer shapes. The reference orientation of the FPD **103** may therefore be expressed by the vector OP.sub.D and the rotation angles about X.sub.D, Y.sub.D, and Z.sub.D when the FPD **103** is housed.
(68) Similarly, the coordinates of the radiation generator are expressed as the vector OP.sub.4. The vector OP.sub.4 is expressed as the sum of four vectors, vector OP.sub.1, vector P.sub.1P.sub.2, vector P.sub.2P.sub.3, and vector P.sub.3P.sub.4.
(69) The vector OP′ is a vector from the reference point O to one end of the second arm **107**. The vector P.sub.1P.sub.2 is a vector from one end of the second arm **107** to the other end (in contact with the first arm **106**) of the second arm **107**. The vector P.sub.2P.sub.3 is a vector from one end of the first arm **106** to the focal point of the tube **115**. The vector P.sub.3P.sub.4 is a vector from the focal point of the tube **115** to the center P.sub.4 of a plane of the aperture from which radiation is emitted. The relationship is expressed as:
{right arrow over (*OP*.sub.4)}={right arrow over (*OP*.sub.1)}+{right arrow over (*P*.sub.1*P*.sub.2)}+{right arrow over (*P*.sub.2*P*.sub.3)}+{right arrow over (*P*.sub.3*P*.sub.4)} Exp. 10
(70) The coordinates of the focal point are expressed as:
{right arrow over (*OP*.sub.3)}={right arrow over (*OP*.sub.1)}+{right arrow over (*P*.sub.1*P*.sub.2)}+{right arrow over (*P*.sub.2*P*.sub.3)} Exp. 11
(71) The vector OP.sub.1 is determined by the dimensions of the base **109**. The vector P.sub.1P.sub.2 is determined by the length (the degree of extension/contraction) and the rotation of the second arm **107**. The vector P.sub.2P.sub.3 is determined by the length (the degree of extension/contraction) and the rotation of the first arm **106**. The vector P.sub.3P.sub.4 is fixed (depends on the position of the focal point of the tube **115** and the dimensions of the aperture **116**).
(72) In other words, in this embodiment, of the vector OP.sub.4, amounts that change at the actual operation by the user are the length, the degree of extension/contraction, orientation, and rotation of the first arm **106**, and the length, orientation, and rotation of the second arm **107**. The initial state of the radiation generator **102** can be calculated from this information.
(73) In transporting the mobile radiography unit **101**, the first arm **106** and the second arm **107** are contracted as much as possible to facilitate transportation. In other words, the arms are in a specific state when the mobile radiography unit **101** is transported. This state is therefore be taken as the reference state of the arms.
(74) Whether the arms are in the reference state can be determined using a known device. The orientation and rotation may be measured using potentiometers disposed at the individual contacts, for example. The extension/contraction of the arms can be measured using a range finder or the like. Alternatively, a mechanical or electrical switch that reacts only when the axes of the arms are in the reference state may be provided.
(75) In this embodiment, the rotations of the radiation generator **102** about the coordinates X.sub.S, Y.sub.S, and Z.sub.S are calculated from the orientation of the first arm **106** and the orientation of the second arm **107**. As described above, the rotation of the first arm **106** about the vector P.sub.2P.sub.3 can be controlled in addition to the direction and magnitude of the vector P.sub.2P.sub.3, and the rotation of the second arm **107** about the vector P.sub.1P.sub.2 can be controlled in addition to the direction and magnitude of the vector P.sub.1P.sub.2. Changing the directions and magnitudes of the arms allows the rotation about the X.sub.S, Y.sub.S, and Z.sub.S to be controlled.
(76) FIG. **4** illustrates a processing procedure of the radiation imaging apparatus **100** in the first embodiment of the present invention. In step **401**, the first control unit **201** determines whether a protocol

for radiation imaging by the user has been selected via the first display **111**.
(77) Immediately after the protocol is selected, in step **402**, the first control unit **201** mounted in the mobile radiography unit **101** determines whether the FPD **103** is housed. The first control unit **201** is connected to the first housing determination unit **112** of the mobile radiography unit in FIG. **2**, so that the first control unit **201** can determine whether the FPD **103** is housed. If the FPD **103** is not properly housed, the reference orientation cannot be accurately calculated. For this reason, a warning is displayed in step **403**, and then the determination whether the FPD **103** is housed is made again in step **402**.
(78) If in step **402** it is determined that the FPD **103** is properly housed, then in step **404** the first control unit **201** generates a trigger and, in response to the trigger, the setting unit **205** in the first control unit **201** sets a reference orientation. In step **402**, it has been determined that the FPD **103** is at a specific position of the mobile radiography unit **101**, and the radiation generator **102** is in an orientation suitable for transportation. For this reason, the reference orientation can be set in step **404**.
(79) In other words, the radiation imaging apparatus **100** of this embodiment can automatically set the reference orientation without a user's specific operation when protocol information for radiation imaging is set, and when the FPD **103** is precent at a specific position (in the FPD housing unit **110**). After the protocol information is set, the user moves the first arm **106** and the second arm **107** but rarely moves the casing **108** and the base **109** to position the radiation generator **102**.
(80) In contrast, the FPD **103** does not move after the protocol information is set until the FPD **103** is drawn out of the FPD housing unit **110**. For this reason, the period after the protocol information is set and while the FPD **103** is in the FPD housing unit **110** is the timing the reference orientation can be set.
(81) In setting the reference orientation, considering both the timing in the imaging procedure and the position allows the reference orientation to be set more accurately. For example, if only the imaging procedure is taken into consideration, in other words, if the reference orientation is set at the selection of the protocol, the reference orientation changes according to the orientation of the FPD **103**, so that the position and angle calculated at the later step depends on the reference orientation.
(82) In contrast, in the case of a system in which the reference orientation is set by determining only the position, for example, whether the FPD housing unit **110** is present, the reference orientation can be set, for example, while the mobile radiography unit **101** is moving, which may decrease the accuracy of the position and angle.
(83) FIGS. **5**A to **5**D FPD **103** illustrate a method of initial setting. FIG. **5**A is a projection view of the FPD **103**, one surface of the casing **108**, the FPD housing unit **110**, and the first housing determination unit **112**, with the FPD **103** housed in the FPD housing unit **110**, in this embodiment. FIGS. **5**B, **5**C, and **5**D illustrate three surfaces in FIG. **5**A.
(84) In this embodiment, the FPD **103**, the FPD housing unit **110**, and one surface of the casing **108** are designed to be parallel with an axis Y.sub.D when the FPD **103** is housed. This design is illustrated in FIGS. **5**B and **5**C. Adopting such a design allows the angular relationship between the FPD **103** and the casing **108** in the initial state (housed state) to be determined. The respective rotation angles (ηD.sub.init and θD.sub.init) about Z-axis and X-axis are set to 0°.
(85) As shown in FIG. **5**D, one surface of the FPD housing unit **110** has an inclination of ϕ.sub.H. The inclination ϕ.sub.H is less than 90°. This structure causes the FPD **103** and one surface of the FPD housing unit **110** to come into contact with each other under gravity, and, the FPD **103** to be parallel to the FPD housing unit **110** when housed. This causes one surface of the casing **108** to be parallel to the axis Y.sub.D. Since the FPD **103** and the FPD housing unit **110** is in contact with each other in the initial state, the rotation angle ϕD.sub.init of the FPD **103** about the axis Y.sub.D in the initial state takes the same value as ϕ.sub.H.
(86) In other words, in this embodiment, the respective rotation angles θD.sub.init. ϕD.sub.init, and ηD.sub.init about the X-axis, the Y-axis, and the Z-axis in the initial state can be calculated from the dimensions of the mobile radiography unit **101**.
(87) For the rotation about the X-axis and the Y-axis of the three axial rotations, the relationship between the orientations and gravity can be calculated from the value from the acceleration sensor. According, the reference orientation may be set using the value from the acceleration sensor. However, only the axial rotation ηD.sub.init about the Z-axis parallel to gravity cannot be obtained by the acceleration sensor, the reference orientation cannot be set from the value from the acceleration sensor. For this reason, providing

a magnetic sensor serving as a detector and calculating an azimuth angle to a magnetic line generated by geomagnetism allows the reference orientation to be set.
(88) The reference orientation of the FPD **103** can be calculated only from the dimensions of the mobile radiography unit **101** as described above. The reference orientation of the radiation generator **102** can be calculated from the orientations of the first arm **106** and the second arm **107** taken at transportation. Thus, using the dimensions of the mobile radiography unit **101** and the values determined from the information on the arms allows the reference orientations of the radiation generator **102** and the FPD **103** to be calculated and set.
(89) In other words, in steps **401** to **404**, the following processes are executed by the setting unit **205** of the radiation imaging apparatus **100**.
(90) The setting unit **205** sets at least one of the reference orientation of the FPD **103** (radiation detector) for calculating the orientation of the FPD **103** and the reference orientation of the radiation generator **102** for calculating the orientation of the radiation generator **102**. This setting is performed by the setting unit **205** in response to detecting the mobile radiography unit **101** equipped with the radiation generator **102** has stood still, with the FPD **103** held. This is the processes executed in steps **401** to **404**.
(91) After the reference orientation is set in step **404**, the orientation acquisition unit **209** starts to acquire and calculate the orientation information on the radiation generator **102** in step **405**. The information to be calculated is the angles and the positions (SID) of the three components. The orientations of the radiation generator **102** and the FPD **103** can be determined from the reference orientations set in step **404** using Exps. 1 to 9.
(92) The SID, if needed, can be expressed as:
(93)

$$\left\| \overrightarrow{P_3P_D} \right\| = \left\| \overrightarrow{OP_D} - \overrightarrow{OP_3} \right\| = \sqrt{(\overrightarrow{OP_D} - \overrightarrow{OP_3})_x^2 + (\overrightarrow{OP_D} - \overrightarrow{OP_3})_y^2 + (\overrightarrow{OP_D} - \overrightarrow{OP_3})_z^2} \quad \text{Exp. 12}$$

where OP.sub.D is the vector of the center of the FPD **103** seen from the origin, OP.sub.3 is a vector of the focal point seen from the origin, P.sub.3P.sub.D is a vector from the focal point to the center of the FPD **103**, and SID is the length of the vector P.sub.3P.sub.D.
(94) In Exp. 12, ( ).sub.x is the magnitude of x-component of the vector. The same applies to y-component and z-component.
(95) Information on the calculated orientations of the radiation generator **102** and the FPD **103** is displayed in step **406**, for example, on the first display **111** or the second display **117**. For example, displaying the angle and the SID on the second display **117** of the radiation generator **102** allows the user to determine whether the current orientations are desired orientations during positioning of the subject.
(96) Next, in step **407**, it is determined whether an instruction to start imaging has been provided. This determination can be made by determining whether an exposure switch (not shown) provided at the mobile radiography unit **101** has been pressed. Step **405** and step **406** are repeated until an imaging start instruction is given, and the user adjusts the orientations so that the radiation field surface of the radiation generator **102** and the FPD **103** face each other while viewing the first display **111** or the second display **117**. When desired orientations are achieved, the user presses the exposure switch to go to step **408**.
(97) If an imaging start instruction is given, then in step **408** the orientation acquisition unit **209** acquires can calculates information on the orientation of the radiation generator **102** immediately before radiation. In step **409**, the orientation information is displayed on the first display **111** or the second display **117**, and then in step **410** the orientation acquisition unit **209** determines the facing relationship between the radiation generator **102** and the FPD **103**.
(98) FIGS. **6**A to **6**C illustrate an example of a method for determining the facing relationship between the radiation generator **102** and the FPD **103**. To determine whether the radiation generator **102** and the FPD **103** face each other, the orientation acquisition unit **209** determines whether the center of the FPD **103** and the center of the radiation field coincide (FIG. **6**A) and determines whether the radiation field surface and the incident surface of the FPD **103** are parallel to each other (FIG. **6**B).
(99) For FIG. **6**A, the difference d.sub.min between the center of the FPD **103** and the center of the

radiation field is expressed as:

(100) $$d_{\min} = \min_{t \geq 0} \text{ .Math. } \overset{\text{.fwdarw.}}{OP_D} - \left(\overset{\text{.fwdarw.}}{OP_3} + t \overset{\text{.fwdarw.}}{e} \right) \text{ .Math. } \quad \text{Exp. 13}$$

where OP.sub.D is the vector of the position of the FPD **103**, and vector OP.sub.3+t×vector e (t is a positive constant and Q is the end point of vector e) is the center of the radiation field.

(101) The determination can be made using Exp. 14 for a threshold T.sub.p of acceptable distances.

(102) $$\begin{cases} \text{If} 0 \leq d_{\min} \leq T_p \text{ thecenterscoincide} \\ \text{If} d_{\min} \geq T_p \text{ ornosolution, thecentersdonotcoincide} \end{cases} \quad \text{Exp. 14}$$

(103) FIG. **6**A illustrates a method for determining whether the center of the FPD **103** and the center of the radiation field coincide.

(104) In other words, whether the positions are proper can be determined by expressing the coordinate P.sub.D and the vector P.sub.3Q as an expression on three-dimensional coordinates, substituting the distance between the point P.sub.D and the half line P.sub.3Q into a known formula, and comparing the distance with the threshold T.sub.p.

(105) The vector e represents the direction in which the radiation from the tube travels. The vector e can be determined from information on the dimensions of the mobile radiography unit **101** and the current orientation of the radiation generator **102** (the current orientation is measured by the first orientation measuring unit **113**). An example of the vector e can be the difference in position between the radiation generator **102** at position P.sub.3 and the aperture **116** at position P.sub.4.

(106) For the determination of the arrangement of the radiation generator **102** and the FPD **103** in FIG. **6**B, the angle ϕ between a normal vector n.sub.1 that aligns with the direction of radiation from the radiation generator **102** and a normal vector n.sub.3 perpendicular to the detection surface of the FPD **103** is determined, and the value cos θ is calculated using Exp. 15.

(107) $$\cos = \frac{\overset{\text{.fwdarw.}}{n_1} \text{ .Math. } \overset{\text{.fwdarw.}}{n_2}}{\text{.Math. } \overset{\text{.fwdarw.}}{n_1} \text{ .Math. } \text{ .Math. } \overset{\text{.fwdarw.}}{n_2} \text{ .Math.}} \quad \text{Exp. 15}$$

where · represents the inner product of the two vectors), and the value cos θ is used for the determination as in Exp. 16.

(108) $$\begin{cases} \text{Ifcos} \leq -1 + T_A \text{, theradiationgeneratorandthe FPDfaceeachother} \\ \text{Ifcos} \geq 1 - T_B \text{, thePFDfacesontheback} \\ \text{Ifotherthantheabove, theradiationgeneratorand theFPDarenotparalleltoeachother} \end{cases} \quad \text{Exp. 16}$$

where T.sub.A and T.sub.B are thresholds representing the tolerance of the angles.

(109) Depending on the imaging procedure, radiation may be applied at an angle. FIG. **6**C illustrates a case in which radiation is applied at an angle to the FPD **103**. For the determination of the arrangement of the radiation generator **102** and the FPD **103** as in FIG. **6**B, the angle ϕ between the normal vector n.sub.1 that aligns with the direction of radiation from the radiation generator **102** and the normal vector n.sub.3 perpendicular to the detection surface of the FPD **103** is determined, and the value cos θ is calculated using Exp. 17.

(110) $$\cos = \frac{\overset{\text{.fwdarw.}}{n_1} \text{ .Math. } \overset{\text{.fwdarw.}}{n_3}}{\text{.Math. } \overset{\text{.fwdarw.}}{n_1} \text{ .Math. } \text{ .Math. } \overset{\text{.fwdarw.}}{n_3} \text{ .Math.}} \quad \text{Exp. 17}$$

where · represents the inner product of the two vectors, and the value cos θ is used for the determination as in Exp. 18.

(111) $$0\ \begin{cases} \text{Ifcos} \quad \le T_d - T_A\text{, theradiationgeneratorandthe FPDarearrangedat}\ a\ \text{predeterminedangle} \\ \text{Ifcos} \quad \ge -T_d + T_B\text{, theFPDfacesontheback} \\ \text{Ifotherthantheabove, theradiationgeneratorand theFPDarenotarrangedatthesameangle} \end{cases} \qquad \text{Exp. 18}$$

where T.sub.d is the inner product of the normal vectors that are set according to the imaging procedure, and T.sub.A and T.sub.B are thresholds representing the tolerance of the angles.

(112) FIGS. **6**B and **6**C illustrate a method for determining the arrangement of the radiation field surface of the radiation generator **102** and the incident surface of the FPD, in which the incident angles of the radiation incident on the FPD **103** differ.

(113) The three angles (angles about axes X.sub.S, Y.sub.S, and Z.sub.S of the radiation generator **102** and the three angles (angles about axes X.sub.D, Y.sub.D, and Z.sub.D) of the FPD **103** are calculated, and the angles are compared, respectively. This allows determining whether the radiation field surface of radiation from the radiation generator **102** and the incident surface of the FPD **103** are arranged according to the imaging procedure.

(114) The orientations of the radiation generator **102** and the FPD **103** are calculated and obtained by the orientation acquisition unit **209** on the basis of the values obtained by the first orientation measuring unit **113** and the second orientation measuring unit **118**, respectively. The normal vectors of the radiation generator **102** and the FPD **103** can be calculated by calculating the normal vectors of the two surfaces in the reference orientations before rotation and then rotating the normal vectors on the basis of angles obtained from the various sensors. The vector e used in FIG. **6**A may also be used as the normal vector n.sub.1 of the radiation generator **102**. A known rotation matrix format may be used in calculating a normal vector after rotation.

(115) Thus, in this embodiment, the orientation acquisition unit **209** acquires the relative relationship between the orientations of the radiation generator **102** and the FPD **103**. If the orientation acquisition unit **209** determines that the centers of the radiation generator **102** and the FPD **103** coincide and the angle between them is within a predetermined range, then the orientation acquisition unit **209** determines whether the radiation generator **102** and the FPD **103** are arranged according to the imaging procedure.

(116) Although the relative relationship between the orientations of the radiation generator **102** and the FPD **103** is obtained by the orientation acquisition unit **209** from the values from the first orientation measuring unit **113** and the second orientation measuring unit **118**, this is illustrative only. For example, the orientation of the radiation generator **102** may be estimated from the positions of the first arm **106** and the second arm **107**, not be calculated from the value from the first orientation measuring unit **113**.

(117) Using at least one of six components, the three components of the position about the three axes and the three components of the angle about the three axes, allows simple determination of facing. For determination of facing only using the angle of rotation about two axes other than rotation about an axis parallel to gravity, for example, only the angles calculated from values obtained by the acceleration sensor may be used. For determination using three-axes rotation, rotation about an axis parallel to gravity calculated from values obtained from an orientation sensor (a magnetic sensor) may be used. Determination of distance may be made using values obtained from a Bluetooth wireless device, for example.

(118) FIG. **4** will be described again hereinbelow. If in step **410** it is determined that the centers of the radiation generator **102** and the FPD **103** do not coincide, or the radiation generator **102** and the FPD **103** are not arranged in a predetermined state, the radiation generator **102** does not emit radiation to the FPD **103**, and the process goes to step **411**.

(119) In step **411**, it is determined whether the FPD **103** is in back placement, and a waning of back placement is displayed in step **412** or a warning of angle mismatch is displayed in step **413**. Separating warnings for center not coincide and placement mismatch (angle mismatch or back placement) allows the user to be given a hint for repositioning.

(120) While this embodiment is configured not to permit radiation, radiation may be permitted, with only a warning displayed. Radiation may be permitted when the button is pressed for a given time, or when the

button is pressed again after a warning is displayed.
(121) If the radiation generator **102** and the FPD **103** face each other, then in step **414** acquisition of information on the orientations of the radiation generator **102** and the FPD **103** ends. Next, in step **415**, the FPD **103** generates an image based on the radiation emitted from the radiation generator **102**.
(122) In step **416**, information on the orientations of the radiation generator **102** and the FPD **103** is added to the image. The addition of orientation information to the image can be performed by, for example, the first control unit **201** in the mobile radiography unit **101**. Examples of a method of addition include common methods, such as addition to the header of the image, embedding into the image itself (for example, decreasing the pixel values of the image area allows embedding of numerical values of orientations into the image), and generation of a dedicated file in which orientation information is recorded. Examples of the orientation information added include the angles and the SIDs of the radiation generator **102** and the FPD **103**.
(123) In step **417**, the acquired and added orientation information is displayed on the first display **111** and the second display **117**. This allows the user to determine whether imaging has succeeded and, if failed (imaging failure), to examine the causes of the failure from the acquired radiological image and the orientation information displayed on the first display **111** of the mobile radiography unit **101**.
(124) In this embodiment, the second orientation measuring unit **118** does not necessarily have to be the acceleration sensor, the geomagnetism sensor, or the angular velocity, but can also be any device that measures the rotation and extension/contraction of the first arm **106** and the second arm **107** and the rotation of the aperture **116**.
(125) For example, a mechanism to control rotation and expansion/contraction can be constituted by a known electrical component, a step motor, and the first control unit **201** of the mobile radiography unit **101**. This allows calculation of the orientations using parameters (orientation control parameters) for use in the control mechanism, instead of calculation of the orientations based on the values measured by the sensor. The orientation control parameters may be back-calculated from the conditions of the mechanism (orientations and the state of motion).
(126) In this embodiment, the distances are derived using an acceleration sensor, but a wireless LAN device, a Bluetooth device, or a UWB device may be used, as described above. As an alternative, a magnetic sensor may be used to derive the distances, in which a magnetic field is artificially generated to identify the position. Aa further alternative, an acceleration sensor, a wireless LAN device, a Bluetooth device, a UWB device, and a magnetic sensor may be combined to further improve the distance measurement accuracy.
(127) Similarly for the angles, by combining angles obtained from a gravity sensor, angles calculated from angular velocities, and azimuth angles calculated from geomagnetism, the measurement accuracy can be further enhanced.
(128) In this embodiment, the reference orientations are set at the selection of a protocol. The reference orientation may be set at the timing when the device for which the reference orientation is to be set stands still on certain coordinates fixed on the ground.
(129) For example, if the mobile radiography unit **101** is moving, the radiation generator **102** and the FPD **103** move together with the mobile radiography unit **101**, so that the radiation generator **102** and the FPD **103** move with respect to certain coordinates fixed on the ground. Similarly, in positioning the radiation generator **102** or the FPD **103**, the radiation generator **102** or the FPD **103** moves with respect to certain coordinates fixed on the ground.
(130) In a general imaging procedure, the period in which the FPD **103** stands still with respect to certain coordinates fixed on the ground is during the period from when the mobile radiography unit **101** is moved near the subject **105** until the FPD **103** is moved to start positioning of the FPD **103**. For this reason, by detecting the processes performed during this period of the processing procedure, the reference orientation can be set, with the FPD **103** standing still with respect to certain coordinates fixed on the ground.
(131) Similarly, the period in which the radiation generator **102** stands still with respect to certain coordinates fixed on the ground is during the period from when the mobile radiography unit **101** is moved near the subject **105** until the radiation generator **102** is moved to start positioning of the radiation generator **102**. For this reason, by detecting the processes performed during this period of the processing

procedure, the reference orientation can be set, with the radiation generator **102** standing still with respect to certain coordinates fixed on the ground.
(132) An example of a specific timing for setting the reference orientations is the instant when the FPD **103** or the radiation generator **102** comes into operation for positioning. In this embodiment, the mobile radiography unit **101** and the FPD **103** include the first housing determination unit **112** and the second housing determination unit **114**, respectively, to be connected by the electrical connectors, as described above. At the start of the positioning of the FPD **103**, the first housing determination unit **112** and the second housing determination unit **114** is disconnected, and the first control unit **201** detects the disconnection and can set the reference orientation.
(133) Similarly for the radiation generator **102**, by disposing electrical components for detecting the motion of the arms, such as potentiometers, and detecting the start of rotation with the first control unit **201**, the reference orientation can be set.
(134) In detecting the start of positioning and setting the reference orientations, the FPD **103** or the radiation generator **102** has already started to move when the setting unit **205** detects a trigger and sets the reference orientations. This causes a slight gap from the standstill on certain coordinates on the ground.
(135) For this reason, the setting unit **205** may hold the output values from the first orientation measuring unit **113** and the second orientation measuring unit **118** for a given length of time. In other words, the output values from the first orientation measuring unit **113** and the second orientation measuring unit **118** immediately before the reference orientations are set are held to use in setting the reference orientations.
(136) The period of time to hold the output may be determined by measuring the time from detection of the start of positioning to setting the reference orientations in advance. In one example, the hold time is about 500 milliseconds.
(137) Another example of the timing of setting the reference orientations is the timing when the standstill of the mobile radiography unit **101** is detected. The standstill of the mobile radiography unit **101** is detected, for example, using a method of detecting the rotation speed of the wheels of the mobile radiography unit **101** or using a position detecting sensor, such as an acceleration sensor or a global positioning system (GPS) sensor, mounted in the mobile radiography unit **101**.
(138) For example, when the rotation speed of the wheels or the output values from the sensor becomes a threshold or less, the first control unit **201** determines that the mobile radiography unit **101** has come to rest. After a certain period of time (for example, 1 second) after the first control unit **201** determines that the mobile radiography unit **101** has come to a standstill, the first control unit **201** generates a trigger and, in response to receiving the trigger, the setting unit **205** sets the reference orientations.
(139) If the mobile radiography unit **101** includes a brake or a locking mechanism, the first control unit **201** may detect whether the mobile radiography unit **101** is at a stop in response to a signal to detect that the brake or the locking mechanism is working for a predetermined period of time.
(140) In this embodiment, the functions of the setting unit **205** and the reference calculation unit **208** are held by the first control unit **201** of the mobile radiography unit **101**, and the function of the orientation acquisition unit **209** is held by the second control unit **203** of the FPD **103**, but this is illustrative only. In one modification, the functions of the setting unit **205** and the reference calculation unit **208** may be held by the second control unit **203** of the FPD **103**, as shown in FIG. **7**. Similarly, the function of the orientation acquisition unit **209** may be held by the first control unit **201** of the mobile radiography unit **101**.
Second Embodiment
(141) In the first embodiment, it is assumed that the mobile radiography unit **101** does not move after the initial state is determined. In a second embodiment, it is assumed that a mobile radiography unit **701** moves after the initial state is determined.
(142) FIG. **8** illustrates a configuration example of this embodiment. The configuration of this embodiment is similar to the configuration of the first embodiment except that a sensor **702**, which is a movement measuring unit for the mobile radiography unit **701**, is mounted on the casing **108** of the mobile radiography unit **701**.
(143) In this embodiment, an acceleration sensor serving as a movement measuring unit is mounted on the casing **108**. Instead, the movement measuring unit may have another configuration. For example, the sensor **702** may be a gyroscope or a magnetic sensor, not the acceleration sensor. The wheels **703** of the

mobile radiography unit **701** often connect to a motor that supplies auxiliary power. Detecting the rotation of the motor allows detecting the rotation of the wheels **703** and determining whether the mobile radiography unit **701** is moving. In another example, a camera or an ultrasonic device may be mounted in the mobile radiography unit **701** to detect the motion.
(144) FIG. **9** is a flowchart of this embodiment. The difference from the first embodiment is that determination of whether the mobile radiography unit **701** stands still or is moving and, if moving, the amount of movement using the sensor **702**.
(145) The setting of the reference orientation in step **404** is performed only when the mobile radiography unit **701** is determined to stand still in step **801**. Whether the mobile radiography unit **701** stands still can be determined, for example, from the value of an acceleration obtained by the sensor **702**. If it is assumed that there is no uniform motion, then it can be determined whether the mobile radiography unit **701** stands still only from the magnitude of the acceleration and the angular velocity.
(146) Furthermore, assuming that only the angular velocity does not change (in the case where the mobile radiography unit **701** does not rotate about the sensor **702**), the motion can be determined solely with the magnitude of the acceleration. A magnetic sensor may be used to examine the change in position, or a combination of an acceleration sensor, a gyroscope, a magnetic sensor, and another sensor may be used to determine whether the mobile radiography unit **701** stands still.
(147) Steps **405** and **406** are the same as those of the first embodiment.
(148) In step **802**, it is determined whether the mobile radiography unit **701** moved between step **801** and **802**. This determination can also be made using the sensor **702**.
(149) If in step **802** it is determined that the mobile radiography unit **701** moved, then in step **803** the reference orientation is corrected or reset based on the measured amount of movement. The reference orientation is corrected by calculating the sum (integration) of the accelerations, velocities, and angular velocities to calculate the displacement (the amount of change in position) and the angles and adding the values to the reference orientation set in step **404**. Alternatively, the reference orientation may be reset by performing the setting of the reference orientation in step **404** again, without using the values set in step **404**, to overwrite the reference orientation.
(150) In step **804**, the reset information on the orientations of the radiation generator **102** and the FPD **103** is displayed on the first display **111** or the second display **117**. Subsequent steps are the same as the steps of the first embodiment.
(151) In this embodiment, when the movement of the mobile radiography unit **701** is detected after the reference orientation is set, the reference orientation is reset based on the calculated amount of movement, but this is illustrative only. For example, a warning message that the mobile radiography unit **701** has moved or a warning message about the reset of the reference orientation may be displayed on the first display **111** or the second display **117** to warn the user.
(152) Specifically, when the movement of the mobile radiography unit **701** is detected after the reference orientation is set, the distance of the movement or numerical information on the coordinates may be displayed as a warning message. Furthermore, a message “Mobile radiography unit moved after reference orientation is set”, “Currently unable to calculate accurate orientation”, or “Return the FPD to holder and then take it out” may be displayed. In this case, the user resets the reference orientation by returning the FPD **103** to the FPD housing unit **110** once and then drawing out the FPD **103** again. After the reset of the reference orientation is completed by a user's operation, a message such as “Reference orientation has been reset” may be displayed.
(153) A warning message may be displayed while the reference orientation is being reset. In this case, until the reset of the reference orientation is completed, moving distance or numerical information on the coordinates may be displayed as a warning message. Alternatively, a message “Mobile radiography unit moved after reference orientation is set”, “Currently unable to calculate accurate orientation”, or “Resetting reference orientation” is displayed. After the reset of the reference orientation is completed, a message “Reference orientation has been reset” may be displayed.
Third Embodiment
(154) In the first and second embodiments, the orientation information on the radiation generator **102** and the FPD **103** are used only for displaying their respective orientations or determination whether to emit radiation. In this embodiment, the radiation generator **102**, the first arm **106**, and the second arm **107** are

equipped with a stepping motor (an example of a movable apparatus) at their movable portions. This allows the mobile radiography unit **101** to set the radiation generator **102** in a desired orientation without user intervention.
(155) The mobile radiography unit **101** of this embodiment is the mobile radiography unit **101** of the first embodiment in which a stepping motor (not shown) is mounted. The first display **111** or the second display **117** also serving as a touch panel are provided with an automatic positioning button. The automatic positioning button may be provided separately from the display. In this embodiment, the mobile radiography unit **101** of the first embodiment is equipped with the stepping motor and the automatic positioning button.
(156) FIG. **10** is a flowchart for the processing of the third embodiment. Steps **701**, **404**, **405**, and **406** are the same as step **801**, **404**, **405**, and **406** of the second embodiment.
(157) In step **901**, the mobile radiography unit **101** determines whether the automatic positioning button has been pressed. If yes, then in step **902** the radiation generator **102** is moved.
(158) FIGS. **11**A to **11**C illustrate the details of a method for moving the radiation generator **102**. The case where the orientation of the radiation generator **102** is determined so that the FPD **103** and the radiation generator **102** face each other will be described.
(159) To check the facing, it is determined whether the center of the FPD **103** and the center of the radiation field surface irradiated with the radiation generator **102** coincide and whether the radiation field surface and the incident surface of the FPD **103** are arranged according to the imaging procedure. Accordingly, the two determinations are needed also for automatic positioning. The procedure for automatic positioning is shown in FIGS. **11**A to **11**C.
(160) Alignment of the center of the FPD **103** and the center of the radiation field will be described with reference to FIG. **11**A.
(161) To align the center of the radiation field surface and the center of the incident surface, the radiation generator **102** is moved so that the position indicated by the sum of the vector OP.sub.D representing the position of the FPD **103** and a vector obtained by multiplying the normal vector of the FPD **103** by a positive real number coincide with the center of the radiation field or the focal point of the tube.
(162) These vectors can be calculated from the values obtained from an acceleration sensor, a gyroscope, or a wireless connection device built in the FPD **103** (see the first embodiment). Accordingly, a variable required to determine the position of the radiation generator **102** is only a positive real number for the normal vector n.sub.2. The positive real number is set based on the tube-focal-point distance (SID). Since the SID varies from imaging to imaging, the SID should be user configurable, or typical conditions for each procedure may be held in advance.
(163) Even if the SID is unknown, the centers can be aligned. In this case, the direction and magnitude of the movement are expressed as a vector from the focal point to a point on a straight line represented by OP.sub.D+t×n.sub.2 (t is a real number). The vector is perpendicular to the normal vector n.sub.2.
(164) To arrange the radiation field surface and the incident surface of the FPD **103** parallel in FIG. **11**B, the orientation of the radiation generator **102** is changed so that the angle θ formed between the vector e indicating the radiation direction and the normal vector n.sub.2 of the FPD **103** becomes approximately 180°. In other words, the position is determined so that Exp. 15 reaches approximately −1.
(165) FIG. **11**C illustrate an imaging procedure for setting the irradiation surface and the incident surface of the FPD **103** at an angle. In this case, the orientation of the radiation generator **102** is changed so that the angle ϕ formed between the vector e indicating the radiation direction and the normal vector n.sub.3 of the FPD **103** reaches a predetermined angle determined by the imaging procedure. In other words, the position is determined so that Exp. 17 reaches approximately T.sub.d.
(166) The vector e varies according to the orientation of the radiation generator **102**. In other words, the vector e can be varied using the stepping motors mounted in the first arm **106**, the second arm **107**, and the radiation generator **102**.
(167) If the automatic positioning button has not been pressed, then in step **407** it is determined whether to start imaging. The subsequent procedure is the same as in the second embodiment.
Fourth Embodiment
(168) In this embodiment, not a mobile radiography unit, but an unmovable floor-mounted apparatus (an apparatus in which one of the radiation generator **102**, the first arm **106**, and the second arm **107** is fixed

to a ceiling or a floor) will be described.
(169) FIG. **12** illustrates an example of the configuration of this embodiment. A characteristic of this embodiment is that a floor-mounted apparatus **1101** is provided instead of the mobile radiography unit **101**. The floor-mounted apparatus **1101**, like the mobile radiography unit **101**, includes a radiation generator **102** and a first arm **106** and a second arm **107** that support the radiation generator **102**.
(170) Another characteristic of this embodiment including the floor-mounted apparatus is that a Bucky's radiographic device **1102** is provided instead of the FPD housing unit **110**. The Bucky's radiographic device **1102** includes a bucky for moving a grid for preventing scattered ray from entering the FPD **103**. The FPD **103** is usually housed in the Bucky's radiographic device **1102** and, in imaging a subject **105**, such as a hand, is taken out of the Bucky's radiographic device **1102** for usage.
(171) Like the mobile radiography unit **101**, the radiation generator **102** and the FPD **103** are each provided with an acceleration sensor, a gyroscope, or a magnetic sensor. If the vector OP.sub.1, which indicates the positional relationship between the Bucky's radiographic device **1102** and the floor-mounted apparatus **1101**, is known in advance, the relationship between the orientations of the radiation generator **102** and the FPD **103** can be calculated. Since the positional relationship between the Bucky's radiographic device **1102** and the floor-mounted apparatus **1101** can also be measured at the installation of the individual apparatuses, the vector OP.sub.1 can be installed as in the first to third embodiments.
(172) Other examples of a device that has a similar function as the Bucky's radiographic device **1102** include an apparatus (a cradle) that can be charged when joined with the FPD **103** and an apparatus (a check-in apparatus), with which the FPD **103** is brought into contact, when the FPD **103** is brought into a radiography room.
(173) While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

Claims

1. A radiation imaging apparatus comprising: a radiation detector configured to detect radiation; a mobile radiography unit configured to house the radiation detector and including a radiation generator; and a setting unit configured to set at least one of a reference orientation of the radiation detector for calculating an orientation of the radiation detector and a reference orientation of the radiation generator for calculating an orientation of the radiation generator, wherein the radiation imaging apparatus is configured to perform radiation imaging to generate an image based on the radiation, and wherein the setting unit performs the setting in response to determining that the mobile radiography unit stands still, with the radiation detector housed in the mobile radiography unit.
2. The radiation imaging apparatus according to claim 1, wherein the setting unit performs the setting during a period from when protocol information for performing the radiation imaging is set by a user to when movement of the radiation detector is started.
3. The radiation imaging apparatus according to claim 1, wherein the setting unit performs the setting during a period from when protocol information for performing the radiation imaging is set by a user to when a change in the orientation of the radiation generator is started.
4. The radiation imaging apparatus according to claim 1, wherein the setting unit calculates at least one of the reference orientation of the radiation detector and the reference orientation of the radiation generator based on dimensions of the radiation imaging apparatus.
5. The radiation imaging apparatus according to claim 1, further comprising: a movement measuring unit configured to measure an amount of movement of the mobile radiography unit, wherein, when the mobile radiography unit moves after the setting, the setting unit corrects or resets the reference orientation of the radiation detector and the reference orientation of the radiation generator based on information from the movement measuring unit.
6. The radiation imaging apparatus according to claim 1, further comprising: a movement measuring unit configured to measure an amount of movement of the mobile radiography unit, wherein, when the mobile radiography unit moves after the setting, the setting unit displays at least one of a message that the mobile

radiography unit has moved and a message about resetting of the reference orientation on a display of the mobile radiography unit based on information from the movement measuring unit.
7. The radiation imaging apparatus according to claim 1, further comprising an orientation acquisition unit configured to acquire relative relationship between the orientation of the radiation generator and the orientation of the radiation detector.
8. The radiation imaging apparatus according to claim 7, further comprising: a first orientation measuring unit configured to measure information on the orientation of the radiation generator; and a second orientation measuring unit configured to measure information on the orientation of the radiation detector, wherein the orientation acquisition unit acquires the relative relationship based on the reference orientation, a value measured by the first orientation measuring unit, and a value measured by the second orientation measuring unit.
9. The radiation imaging apparatus according to claim 8, wherein the orientation acquisition unit acquires the orientation of the radiation generator from the reference orientation and the value measured by the first orientation measuring unit, acquires the orientation of the radiation detector from the reference orientation and the value measured by the second orientation measuring unit, and determines whether the radiation generator and the radiation detector face each other from the orientation of the radiation generator and the orientation of the radiation detector.
10. The radiation imaging apparatus according to claim 9, wherein the orientation acquisition unit acquires the relative relationship from an angle formed between a first normal vector and a second normal vector, the first normal vector being aligned with a direction of the radiation from the radiation generator and being based on the reference orientation and the value measured by the first orientation measuring unit, the second normal vector being perpendicular to a detector plane of the radiation detector and being based on the reference orientation and the value measured by the second orientation measuring unit.
11. The radiation imaging apparatus according to claim 10, wherein the orientation acquisition unit determines whether the relative relationship is in a range predetermined according to an imaging procedure from the angle formed between the first normal vector and the second normal vector.
12. The radiation imaging apparatus according to claim 11, wherein the range predetermined according to the imaging procedure is any one of whether the detector plane of the radiation detector faces the direction of radiation from the radiation generator, whether a back of the detector plane of the radiation detector faces the direction of radiation from the radiation generator, and whether the radiation generator and the radiation detector are arranged at a predetermined angle.
13. A radiation imaging apparatus comprising: a radiation detector configured to detect radiation; a floor or ceiling-mounted apparatus including a radiation generator and an arm configured to move the radiation generator; and a setting unit configured to set at least one of a reference orientation of the radiation detector for calculating an orientation of the radiation detector and a reference orientation of the radiation generator for calculating an orientation of the radiation generator, wherein the radiation imaging apparatus is configured to perform radiation imaging to generate an image based on the radiation, and wherein the setting unit performs the setting in response to determining that the radiation detector unit stands still.
14. The radiation imaging apparatus according to claim 13, wherein the floor or ceiling-mounted apparatus comprises a Bucky's radiographic device including a bucky for moving a grid for preventing incidence of scattering rays, and wherein the setting unit performs the setting when the radiation detector is installed at the Bucky's radiographic device.
15. The radiation imaging apparatus according to claim 13, wherein the floor or ceiling-mounted apparatus is a cradle for charging the radiation detector, and wherein the setting unit performs the setting when the radiation detector is installed at the cradle.
16. A radiation imaging apparatus comprising: a radiation detector configured to detect radiation; a mobile radiography unit configured to house the radiation detector and including a radiation generator; an orientation acquisition unit configured to acquire relative relationship between an orientation of the radiation generator and an orientation of the radiation detector; a first orientation measuring unit configured to measure information on the orientation of the radiation generator; and a second orientation measuring unit configured to measure information on the orientation of the radiation detector, wherein the radiation imaging apparatus is configured to perform radiation imaging to generate an image based on the radiation, and wherein the orientation acquisition unit acquires the relative relationship from an angle

formed between a first normal vector and a second normal vector, the first normal vector being aligned with a direction of the radiation from the radiation generator and being based on a value measured by the first orientation measuring unit, the second normal vector being perpendicular to a detector plane of the radiation detector and being based on a value measured by the second orientation measuring unit.

17. The radiation imaging apparatus according to claim 16, wherein the mobile radiography unit includes a display, and wherein the orientation acquisition unit displays information on arrangement of the radiation generator and the radiation detector on the display based on the relative relationship.

18. The radiation imaging apparatus according to claim 16, wherein the setting unit holds output values from the first orientation measuring unit and the second orientation measuring unit immediately before the setting is performed and performs the setting based on the immediately preceding output values.

19. The radiation imaging apparatus according to claim 16, wherein the first orientation measuring unit includes at least one of an acceleration sensor, an angular velocity sensor, and a geomagnetic sensor.

20. The radiation imaging apparatus according to claim 16, wherein the second orientation measuring unit includes at least one of an acceleration sensor, an angular velocity sensor, and a geomagnetic sensor.

21. The radiation imaging apparatus according to claim 16, further comprising: a movable apparatus configured to change the orientation of the radiation generator, wherein the movable apparatus changes the orientation of the radiation generator based on the relative relationship.

22. A radiation detector configured to be housed in a mobile radiography unit including a radiation generator and configured to detect radiation to perform radiation imaging to generate an image based on the radiation, the radiation detector comprising: a setting unit configured to set at least one of a reference orientation of the radiation detector for calculating an orientation of the radiation detector and a reference orientation of the radiation generator for calculating an orientation of the radiation generator, wherein the setting unit performs the setting in response to determining that the mobile radiography unit stands still, with the radiation detector housed in the mobile radiography unit.

23. A radiation detector configured to be housed in a mobile radiography unit including a radiation generator and configured to detect radiation to perform radiation imaging to generate an image based on the radiation, the radiation detector comprising: an orientation measuring unit configured to measure information on the orientation of the radiation generator; and an orientation acquisition unit configured to acquire relative relationship between the orientation of the radiation generator and the orientation of the radiation detector, wherein the orientation acquisition unit acquires the relative relationship from an angle formed between a first normal vector and a second normal vector, the first normal vector being aligned with a direction of the radiation from the radiation generator and being based on a value measured by an orientation measuring unit different from the orientation measuring unit, the second normal vector being perpendicular to a detector plane of the radiation detector and being based on a value measured by the orientation measuring unit.

24. A control apparatus configured to control a radiation imaging apparatus including a radiation detector configured to detect radiation and a mobile radiography unit configured to house the radiation detector and including a radiation generator, the radiation imaging apparatus being configured to perform radiation imaging to generate an image based on the radiation, wherein the control apparatus sets at least one of a reference orientation of the radiation detector for calculating an orientation of the radiation detector and a reference orientation of the radiation generator for calculating an orientation of the radiation generator in response to determining that the mobile radiography unit stands still, with the radiation detector housed in the mobile radiography unit.

25. A control apparatus configured to control a radiation imaging apparatus including a radiation detector configured to detect radiation and a mobile radiography unit configured to house the radiation detector and including a radiation generator, the radiation imaging apparatus being configured to perform radiation imaging to generate an image based on the radiation, wherein the control apparatus acquires relative relationship between an orientation of the radiation detector and an orientation of the radiation generator from an angle formed between a first normal vector and a second normal vector, the first normal vector being aligned with a direction of the radiation from the radiation generator and being based on a value calculated by a first orientation measuring unit provided at the radiation generator and measuring information on the orientation of the radiation generator, the second normal vector being perpendicular to a detector plane of the radiation detector and being based on a value calculated by a second orientation

measuring unit provided at the radiation detector and measuring information on the orientation of the radiation detector.

26. A method for controlling a radiation imaging apparatus including a radiation detector configured to detect radiation and a mobile radiography unit configured to house the radiation detector and including a radiation generator, the radiation imaging apparatus being configured to perform radiation imaging to generate an image based on the radiation, the method comprising: setting at least one of a reference orientation of the radiation detector for calculating an orientation of the radiation detector and a reference orientation of the radiation generator for calculating an orientation of the radiation generator in response to determining that the mobile radiography unit stands still, with the radiation detector housed in the mobile radiography unit.

27. A method for controlling a radiation imaging apparatus including a radiation detector configured to detect radiation and a mobile radiography unit configured to house the radiation detector and including a radiation generator, the radiation imaging apparatus being configured to perform radiation imaging to generate an image based on the radiation, the method comprising: acquiring relative relationship between an orientation of the radiation detector and an orientation of the radiation generator from an angle formed between a first normal vector and a second normal vector, the first normal vector being aligned with a direction of the radiation from the radiation generator and being based on a value calculated by a first orientation measuring unit provided at the radiation generator and measuring information on the orientation of the radiation generator, the second normal vector being perpendicular to a detector plane of the radiation detector and being based on a value calculated by a second orientation measuring unit provided at the radiation detector and measuring information on the orientation of the radiation detector.

28. A non-transitory computer-readable medium storing a program for causing a computer to execute the control method according to claim 26.